Practical - 8

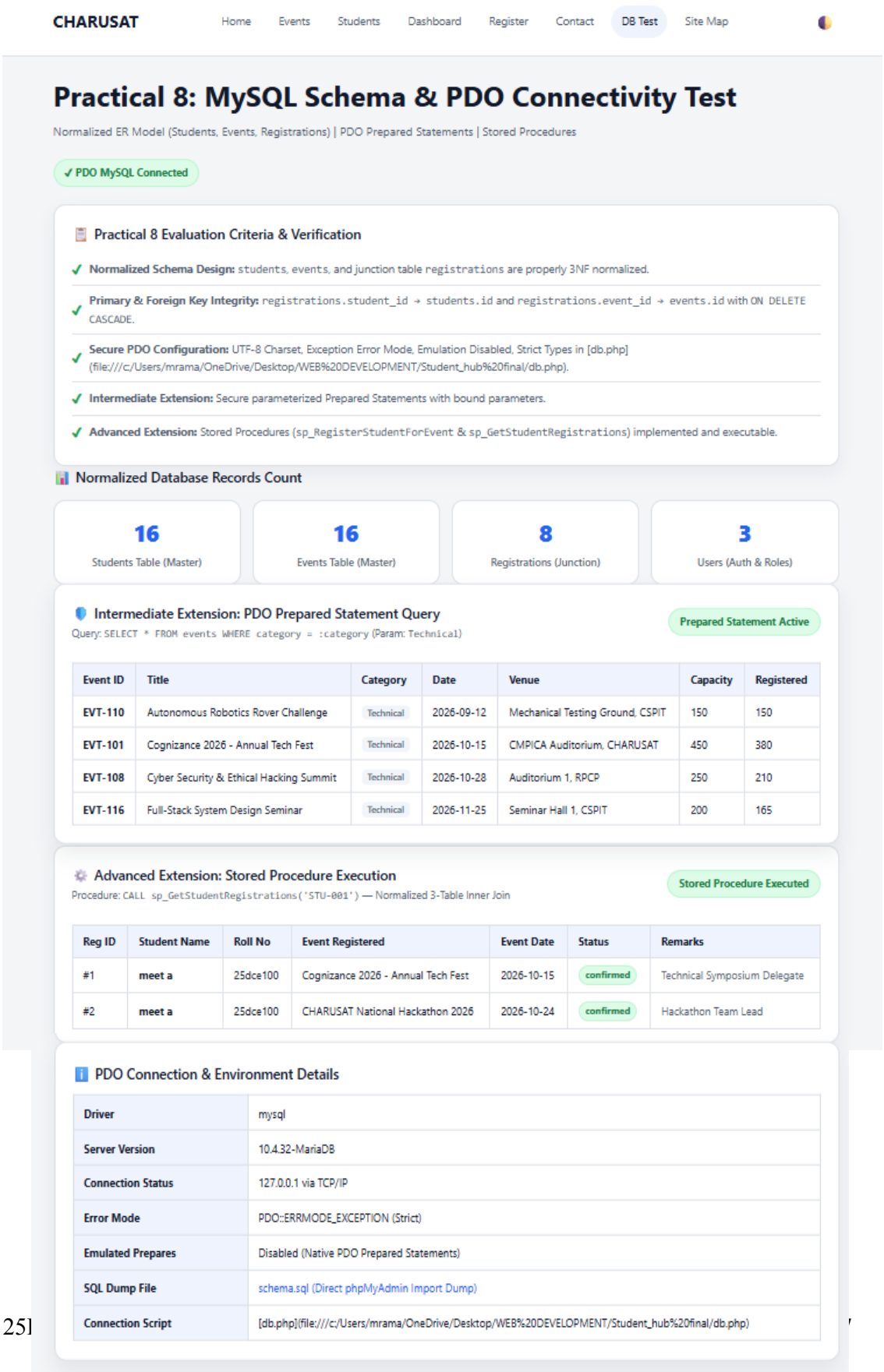
Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements
Problem Definition: Design the database for Student Hub with students, events and registrations. Create PHP database connection using MySQL statements.

Theory :

1. Is the schema normalized properly?
Yes, the schema is normalized up to the Third Normal Form (3NF). Student details, event details and registration details are stored in separate tables (students, events and registrations), so there is no repeated data. The registrations table connects students and events (many-to-many relationship), and every non-key column depends only on the primary key of its table.

2. Are primary keys and foreign keys correctly defined?
Yes. Each table has an AUTO_INCREMENT primary key (student_id, event_id and registration_id). The registrations table has student_id and event_id as foreign keys that refer to the students and events tables. A UNIQUE constraint on (student_id, event_id) stops the same student from registering twice for the same event.

3. Is the database connection handled securcly?
Yes. The connection is written once in db.php using PHP PDO inside a try-catch block with exception error mode. Prepared statements are used so that SQL injection is avoided, and a simple error message is shown to the user instead of the real database error details.


CHARUSAT
Home
Events
Students
Dashboard
Register
Contact
DB Test
Site Map
Practical 8: MySQL Schema & PDO Connectivity Test
Normalized ER Model (Students, Events, Registrations) | PDO Prepared Statements | Stored Procedures
✓ PDO MySQL Connected
Practical 8 Evaluation Criteria & Verification
✓ Normalized Schema Design: students, events, and junction table registrations are properly 3NF normalized.
✓ Primary & Foreign Key Integrity: registrations.student_id → students.id and registrations.event_id → events.id with ON DELETE CASCADE.
✓ Secure PDO Configuration: UTF-8 Charset, Exception Error Mode, Emulation Disabled, Strict Types in [db.php] (file:///c:/Users/mrama/OneDrive/Desktop/WEB%20DEVELOPMENT/Student_hub%20final/db.php).
✓ Intermediate Extension: Secure parameterized Prepared Statements with bound parameters.
✓ Advanced Extension: Stored Procedures (sp_RegisterStudentForEvent & sp_GetStudentRegistrations) implemented and executable.
Normalized Database Records Count
16
Students Table (Master)
16
Events Table (Master)
8
Registrations (Junction)
3
Users (Auth & Roles)
Intermediate Extension: PDO Prepared Statement Query
Query: SELECT * FROM events WHERE category = :category (Param: Technical)
Prepared Statement Active
Event ID
Title
Category
Date
Venue
Capacity
Registered
EVT-110
Autonomous Robotics Rover Challenge
Technical
2026-09-12
Mechanical Testing Ground, CSPIT
150
150
EVT-101
Cognizance 2026 - Annual Tech Fest
Technical
2026-10-15
CMPICA Auditorium, CHARUSAT
450
380
EVT-108
Cyber Security & Ethical Hacking Summit
Technical
2026-10-28
Auditorium 1, RPCP
250
210
EVT-116
Full-Stack System Design Seminar
Technical
2026-11-25
Seminar Hall 1, CSPIT
200
165
Advanced Extension: Stored Procedure Execution
Procedure: CALL sp_GetStudentRegistrations('STU-001') — Normalized 3-Table Inner Join
Stored Procedure Executed
Reg ID
Student Name
Roll No
Event Registered
Event Date
Status
Remarks
#1
meet a
25dce100
Cognizance 2026 - Annual Tech Fest
2026-10-15
confirmed
Technical Symposium Delegate
#2
meet a
25dce100
CHARUSAT National Hackathon 2026
2026-10-24
confirmed
Hackathon Team Lead
PDO Connection & Environment Details
Driver
mysql
Server Version
10.4.32-MariaDB
Connection Status
127.0.0.1 via TCP/IP
Error Mode
PDO::ERRMODE_EXCEPTION (Strict)
Emulated Prepares
Disabled (Native PDO Prepared Statements)
SQL Dump File
schema.sql (Direct phpMyAdmin Import Dump)
Connection Script
[db.php](file:///c:/Users/mrama/OneDrive/Desktop/WEB%20DEVELOPMENT/Student_hub%20final/db.php)

Repository Link :-
https://github.com/meet-3834/Student-Hub/tree/main/FINAL%20Submission

H. Learning Outcomes:
I am able to design and connect a normalized database securely using PHP.

Faculty Remarks: (Keep this blank for faculty)